

Text Processing

1. Raw Ingestion

JSONL / CSV sources

2. Deduplication

URL / Hash collision

3. Text Cleaning

regex, markdown

4. Token Truncation

enforce max length

drop invalid text

Vision Validation

8. Resize & Pad

conform to 1024px

7. Visual Curation

blur, resolution

6. Format Verify

JPEG/PNG headers

5. Image Download

fetch URLs, timeout

drop corrupt/blur

Structuring & Export

9. Structuring

map to conversation

10. Tokenization

Qwen2.5 chat template

11. Packing

batch chunks

dataset export

Final Dataset

HuggingFace format

Design rationale — The pipeline is strictly sequential. Text cleaning occurs before image downloading to save bandwidth by immediately dropping malformed textual records. Visual curation guarantees that the model receives only crisp, adequately sized imagery, which is critical for the visual token representations in the Qwen architecture.